Can Large Language Models Improve the Adversarial Robustness of Graph Neural Networks? (KDD ’25)

Chloe Kim
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Problem:

- LLMs can improve GNNs' performance by enriching node features.
- Although LLMs can improve the robustness of GNNs, there's still a decrease in accuracy.

Clean Graph Adversarial Robustness

The authors tested on the Cora and PubMed datasets, applied Mettack with a 20% perturbation rate to generate a topology attack, and compared results against seven representative baselines (shown in the figure).

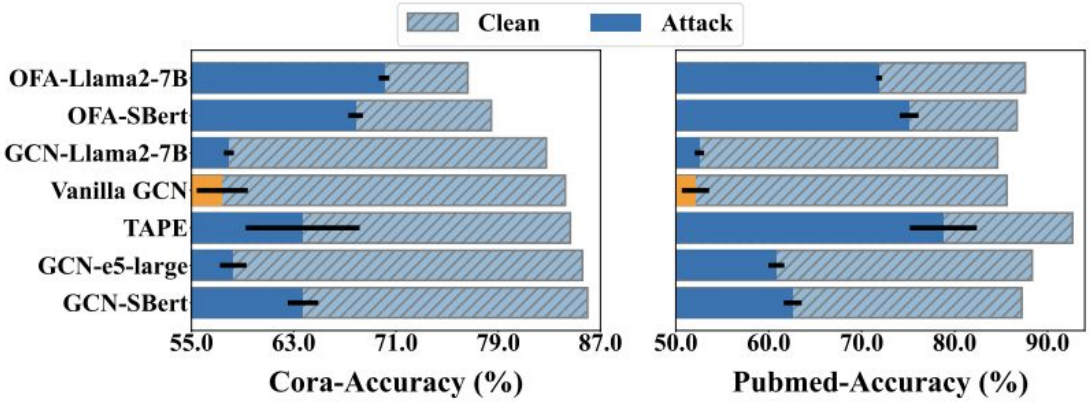


Figure 1: The accuracy of different GNNs combining LLMs/LMs against Mettack with a 20% perturbation rate.

Problem:

- “How can we use LLMs to make GNNs robust to topology attacks, beyond just boosting clean accuracy, by detecting and correcting malicious edge edits?”

Goal: Defending against stronger poisoning attacks

- Poisoning attacks can lead to an extremely low model performance by directly modifying the training data.
- They focus on 2 topology attacks:
  - **Targeted attacks** where attackers aim to mislead the model's prediction on specific nodes by manipulating the adjacent edges.
  - **Non-targeted attacks** where attackers aim to degrade the overall performance of GNNs but do not care which node is being targeted.

Solution: LLM4RGNN

- LLM4RGNN is an LLM-based robust graph structure inference framework.

Method

LLM4RGNN involves 3 parts:

a. Instruction tuning a local LLM
b. Training an LM-based edge predictor
c. Purifying the graph structure

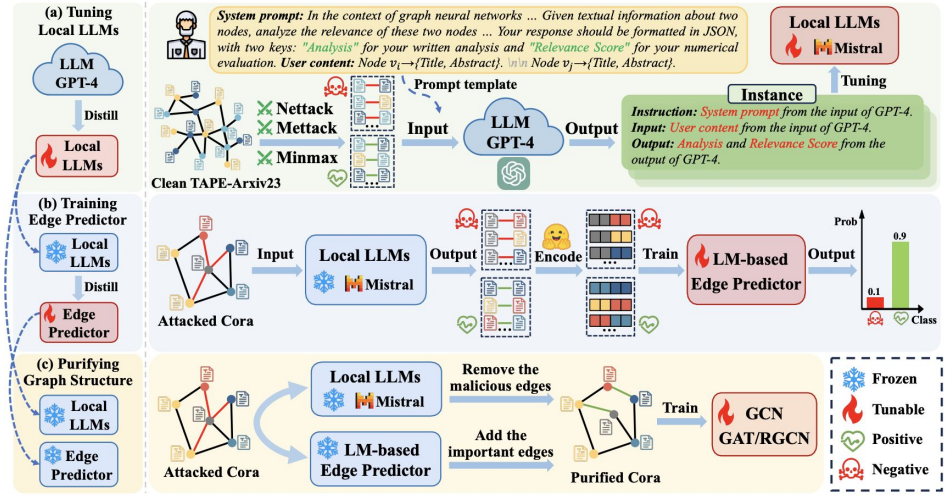


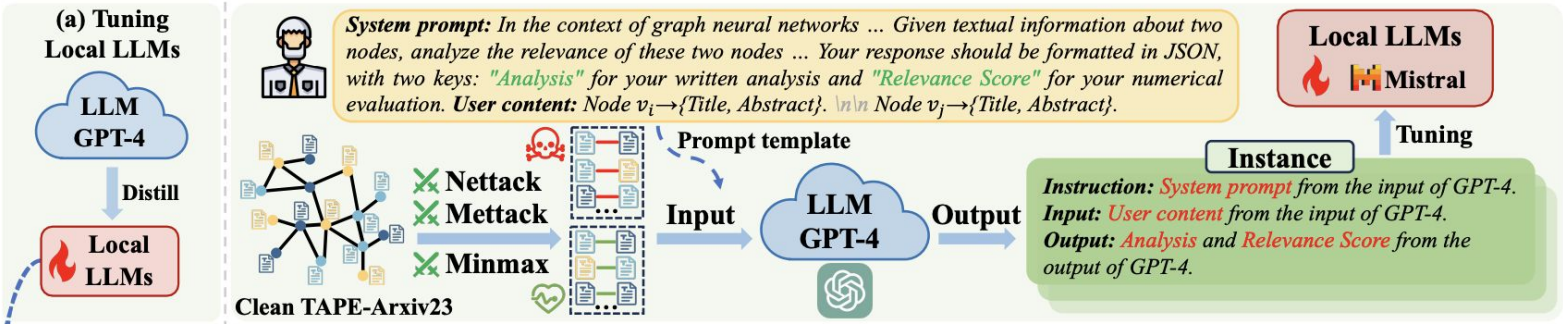
Figure 2: Overall framework of LLM4RGNN.

Instruction tuning

- Instruction dataset which consists of: Instruction, Input, and Output.
- Key = how to construct an effective instruction dataset for fine-tuning an LLM to identify the malicious edges.

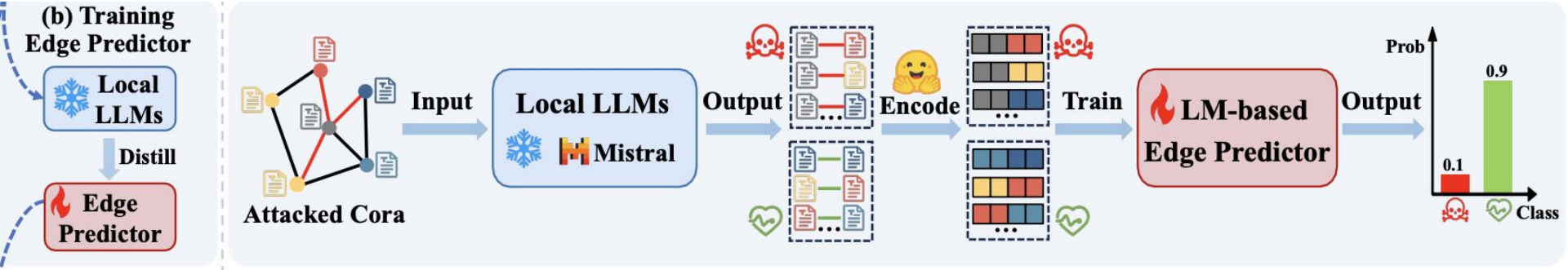
System prompt: In the context of graph neural networks, attackers manipulate models by adding irrelevant edges or removing relevant ones, leading to incorrect predictions. Your role is crucial in defending against such attacks by evaluating the relevance between pairs of nodes, which will help in identifying and removing the irrelevant edges to mitigate the impact of adversarial attacks on graph-based models. Given textual information about two nodes, analyze the relevance of these two nodes. Provide a concise analysis(approximately 100 words) and assign an integer relevance score from 1 to 6, where 1 indicates completely irrelevant and 6 indicates directly relevant. Your response should be formatted in JSON, with two keys: "Analysis" for your written analysis and "Relevance Score" for your numerical evaluation.
User content: Node v_i→{Title, Abstract}.\n\nNode v_j→{Title, Abstract}.

Analysis: {Analysis of predicted results}.
Relevance Score: {Predicted integer scores from 1-6}.



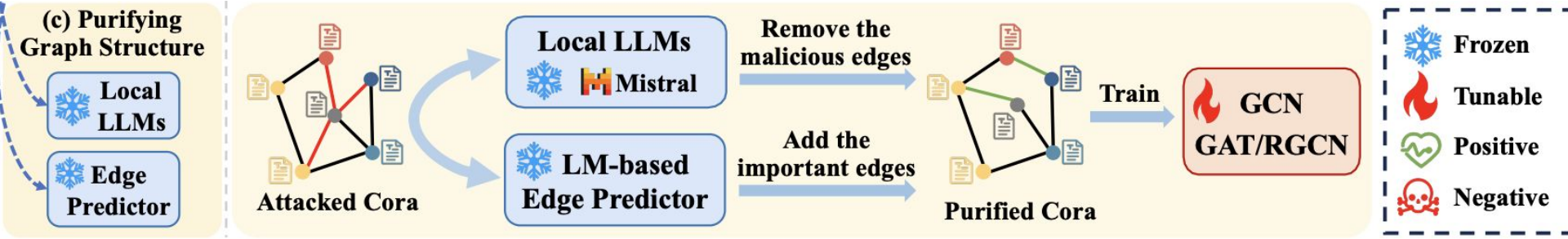
Edge predictor training

- Need to find and add important edges that do not exist in the attacked graph structure.
- Local LLM could do this, but it’s very time consuming.
- SBert as text encoder -> train a lightweight MLP
- The ML-based EP gives each missing pair a probability.



Graph purification

- New edges are added in the attacked graph structure.
- Simultaneously, remove the malicious edges.



Experiments

- **Dataset**:
  - Cora, Citeseer, Pubmed, and OGBN-Arxiv
  - OGBN-Products

- **Baseline**: LLM4RGNN is a general LLM-based framework
  - GCN and 3 robust GNNs (GAT, RGCN, and Simp-GCN) as baselines.
  - In absence of robust GNN frameworks such as ProGNN, STABLE, HANGquad, and GraphEdit

Main Results

How well does LLM4RGNN hold against…

1. Mettack
2. DICE
3. Nettack
4. Adaptive Attack

Main Result Against Mettack

Non-targeted

Table 1: Node classification accuracy ($\% \pm \sigma$) under non-targeted attack (Mettack). Bolded results indicate improved performance.

Dataset	Ptb Rate	GCN		GAT		RGCN		SimP-GCN	
		Vanilla	LLM4RGNN	Vanilla	LLM4RGNN	Vanilla	LLM4RGNN	Vanilla	LLM4RGNN
Cora	0%	$84.25_{\pm 0.36}$	$84.13_{\pm 0.33}$	$84.62_{\pm 0.47}$	$84.61_{\pm 0.39}$	$84.61_{\pm 0.46}$	$84.50_{\pm 0.46}$	$84.70_{\pm 0.62}$	$84.46_{\pm 0.67}$
	5%	$75.62_{\pm 1.70}$	**$81.76_{\pm 0.69}$**	$79.82_{\pm 0.52}$	**$81.22_{\pm 0.78}$**	$76.65_{\pm 0.82}$	**$82.19_{\pm 0.59}$**	$79.36_{\pm 0.99}$	**$81.78_{\pm 0.51}$**
	10%	$70.72_{\pm 2.86}$	**$81.80_{\pm 0.76}$**	$75.33_{\pm 1.17}$	**$81.86_{\pm 0.61}$**	$69.62_{\pm 1.09}$	**$82.14_{\pm 0.60}$**	$77.00_{\pm 1.34}$	**$82.19_{\pm 0.75}$**
	20%	$57.41_{\pm 2.00}$	**$81.41_{\pm 0.77}$**	$63.17_{\pm 1.82}$	**$81.00_{\pm 0.98}$**	$59.27_{\pm 0.81}$	**$81.39_{\pm 0.44}$**	$76.04_{\pm 1.66}$	**$81.20_{\pm 0.63}$**
Citeseer	0%	$73.38_{\pm 0.72}$	**$74.20_{\pm 0.56}$**	$73.95_{\pm 0.45}$	$73.72_{\pm 1.07}$	$74.45_{\pm 0.31}$	$74.07_{\pm 0.90}$	$72.71_{\pm 1.14}$	**$73.30_{\pm 0.62}$**
	5%	$69.69_{\pm 0.59}$	**$73.94_{\pm 0.56}$**	$71.37_{\pm 0.73}$	**$73.22_{\pm 1.08}$**	$72.30_{\pm 0.25}$	**$73.61_{\pm 0.65}$**	$71.07_{\pm 0.93}$	**$73.48_{\pm 0.58}$**
	10%	$65.75_{\pm 0.94}$	**$73.62_{\pm 0.39}$**	$68.40_{\pm 1.14}$	**$73.21_{\pm 1.16}$**	$69.36_{\pm 0.40}$	**$73.41_{\pm 0.90}$**	$70.55_{\pm 0.55}$	**$72.84_{\pm 0.50}$**
	20%	$58.72_{\pm 1.00}$	**$74.12_{\pm 0.85}$**	$61.92_{\pm 1.76}$	**$73.94_{\pm 0.67}$**	$62.79_{\pm 0.60}$	**$74.04_{\pm 0.70}$**	$69.54_{\pm 0.78}$	**$73.70_{\pm 0.64}$**
Pubmed	0%	$85.62_{\pm 0.10}$	**$86.21_{\pm 0.13}$**	$85.14_{\pm 0.11}$	$85.10_{\pm 0.09}$	$85.84_{\pm 0.10}$	**$86.35_{\pm 0.10}$**	$87.26_{\pm 0.08}$	**$87.53_{\pm 0.14}$**
	5%	$73.27_{\pm 1.15}$	**$85.27_{\pm 0.26}$**	$80.87_{\pm 0.91}$	**$84.45_{\pm 0.23}$**	$81.61_{\pm 0.20}$	**$85.93_{\pm 0.10}$**	$85.90_{\pm 0.20}$	**$86.08_{\pm 0.27}$**
	10%	$67.54_{\pm 0.42}$	**$84.95_{\pm 0.17}$**	$69.94_{\pm 4.60}$	**$84.88_{\pm 0.19}$**	$69.55_{\pm 0.40}$	**$85.79_{\pm 0.22}$**	$85.71_{\pm 0.11}$	**$86.06_{\pm 0.13}$**
	20%	$52.12_{\pm 1.47}$	**$84.99_{\pm 0.32}$**	$53.07_{\pm 0.69}$	**$85.20_{\pm 0.30}$**	$48.47_{\pm 0.71}$	**$85.99_{\pm 0.21}$**	$85.70_{\pm 0.15}$	**$86.07_{\pm 0.24}$**
Arxiv	0%	$67.01_{\pm 0.08}$	**$68.16_{\pm 0.36}$**	$65.02_{\pm 0.37}$	**$67.43_{\pm 0.49}$**	$65.53_{\pm 0.38}$	**$67.53_{\pm 0.38}$**	$65.13_{\pm 0.31}$	**$66.20_{\pm 1.16}$**
	5%	$50.51_{\pm 0.29}$	**$68.86_{\pm 0.41}$**	$54.68_{\pm 0.57}$	**$68.46_{\pm 0.47}$**	$52.35_{\pm 0.09}$	**$68.71_{\pm 0.43}$**	$51.07_{\pm 3.87}$	**$67.88_{\pm 0.47}$**
	10%	$42.91_{\pm 0.81}$	**$68.65_{\pm 0.43}$**	$49.18_{\pm 0.54}$	**$68.56_{\pm 0.55}$**	$44.75_{\pm 0.36}$	**$68.54_{\pm 0.70}$**	$46.52_{\pm 6.76}$	**$66.47_{\pm 2.54}$**
	20%	$33.96_{\pm 0.46}$	**$69.17_{\pm 0.43}$**	$34.24_{\pm 2.12}$	**$68.86_{\pm 0.54}$**	$31.69_{\pm 0.63}$	**$69.00_{\pm 0.59}$**	$37.31_{\pm 7.59}$	**$68.15_{\pm 0.65}$**
Products	0%	$79.86_{\pm 0.15}$	$79.04_{\pm 0.42}$	$78.75_{\pm 0.26}$	$77.76_{\pm 0.62}$	$78.45_{\pm 0.25}$	$77.83_{\pm 0.48}$	$75.87_{\pm 0.36}$	$75.84_{\pm 0.29}$
	5%	$66.62_{\pm 0.56}$	**$76.34_{\pm 0.39}$**	$76.41_{\pm 0.60}$	**$76.70_{\pm 0.68}$**	$71.48_{\pm 0.35}$	**$75.20_{\pm 0.25}$**	$64.84_{\pm 0.60}$	**$72.67_{\pm 0.66}$**
	10%	$63.31_{\pm 0.52}$	**$75.80_{\pm 0.24}$**	$74.13_{\pm 0.37}$	**$75.48_{\pm 0.75}$**	$68.98_{\pm 0.41}$	**$74.75_{\pm 0.38}$**	$58.59_{\pm 1.97}$	**$71.80_{\pm 0.48}$**
	20%	$57.56_{\pm 0.64}$	**$76.57_{\pm 0.46}$**	$70.25_{\pm 0.82}$	**$74.98_{\pm 0.53}$**	$64.81_{\pm 0.31}$	**$74.32_{\pm 0.38}$**	$50.36_{\pm 0.96}$	**$73.71_{\pm 0.81}$**

Table 3: Node classification accuracy ($\% \pm \sigma$) under non-targeted attack (Mettack). The best results are in bold. OOT means that the result could not be obtained in 15 days.

Dataset	Ptb Rate	Pro-GNN	STABLE	HANG-quad	GraphEdit	LLM4RGNN
Cora	0%	$80.85_{\pm 0.44}$	**$85.09_{\pm 0.21}$**	$79.41_{\pm 0.55}$	$76.35_{\pm 0.38}$	$84.13_{\pm 0.33}$
	5%	$79.81_{\pm 0.45}$	$80.53_{\pm 1.13}$	$78.81_{\pm 1.02}$	$74.97_{\pm 0.95}$	**$81.76_{\pm 0.69}$**
	10%	$78.57_{\pm 0.97}$	$79.53_{\pm 0.52}$	$78.15_{\pm 1.14}$	$74.87_{\pm 0.75}$	**$81.80_{\pm 0.76}$**
	20%	$76.07_{\pm 0.57}$	$78.70_{\pm 0.89}$	$74.90_{\pm 0.65}$	$73.82_{\pm 0.48}$	**$81.41_{\pm 0.77}$**
Citeseer	0%	$71.11_{\pm 0.45}$	$72.51_{\pm 1.37}$	$71.18_{\pm 0.64}$	$72.60_{\pm 0.57}$	**$74.20_{\pm 0.56}$**
	5%	$69.68_{\pm 0.53}$	$71.23_{\pm 1.35}$	$71.16_{\pm 0.71}$	$71.40_{\pm 0.72}$	**$73.94_{\pm 0.56}$**
	10%	$68.73_{\pm 0.79}$	$70.47_{\pm 1.47}$	$70.84_{\pm 1.00}$	$71.35_{\pm 0.98}$	**$73.62_{\pm 0.39}$**
	20%	$68.37_{\pm 0.84}$	$67.91_{\pm 1.98}$	$70.00_{\pm 1.11}$	$69.44_{\pm 0.86}$	**$74.12_{\pm 0.85}$**
Pubmed	0%	OOT	$84.07_{\pm 0.17}$	$84.98_{\pm 0.13}$	$85.38_{\pm 0.07}$	**$86.21_{\pm 0.13}$**
	5%	OOT	$79.41_{\pm 0.86}$	$84.97_{\pm 0.16}$	$84.77_{\pm 0.09}$	**$85.27_{\pm 0.26}$**
	10%	OOT	$77.65_{\pm 0.25}$	$84.88_{\pm 0.19}$	$83.41_{\pm 0.19}$	**$84.95_{\pm 0.17}$**
	20%	OOT	$72.51_{\pm 1.05}$	$84.94_{\pm 0.19}$	$82.34_{\pm 0.27}$	**$84.99_{\pm 0.32}$**
Arxiv	0%	OOT	$66.70_{\pm 0.19}$	$67.85_{\pm 0.17}$	-	**$68.16_{\pm 0.36}$**
	5%	OOT	$61.40_{\pm 0.50}$	$63.53_{\pm 0.42}$	-	**$68.86_{\pm 0.41}$**
	10%	OOT	$59.37_{\pm 0.40}$	$57.49_{\pm 0.76}$	-	**$68.65_{\pm 0.43}$**
	20%	OOT	$58.24_{\pm 0.60}$	$41.61_{\pm 1.28}$	-	**$69.17_{\pm 0.43}$**
Products	0%	OOT	$79.23_{\pm 0.33}$	**$80.61_{\pm 0.18}$**	-	$79.04_{\pm 0.42}$
	5%	OOT	$76.24_{\pm 0.44}$	$72.41_{\pm 0.66}$	-	**$76.34_{\pm 0.39}$**
	10%	OOT	$74.38_{\pm 0.40}$	$69.88_{\pm 0.86}$	-	**$75.80_{\pm 0.24}$**
	20%	OOT	$72.21_{\pm 0.40}$	$62.24_{\pm 0.28}$	-	**$76.57_{\pm 0.46}$**

Main Result Against DICE

Non-targeted

Table 2: Node classification accuracy (% ± σ) under non-targeted attack (DICE). Bolded results indicate improved performance.

Dataset	Ptb Rate	GCN		GAT		RGCN		SimP-GCN	
		Vanilla	LLM4RGNN	Vanilla	LLM4RGNN	Vanilla	LLM4RGNN	Vanilla	LLM4RGNN
Cora	10%	$81.38_{\pm0.31}$	**$81.84_{\pm0.71}$**	$81.52_{\pm0.49}$	**$82.12_{\pm0.71}$**	$81.71_{\pm0.52}$	**$82.14_{\pm0.52}$**	$82.29_{\pm0.55}$	**$82.35_{\pm0.65}$**
	20%	$78.38_{\pm0.51}$	**$80.96_{\pm0.61}$**	$78.51_{\pm0.35}$	**$81.02_{\pm1.00}$**	$78.58_{\pm0.55}$	**$81.52_{\pm0.88}$**	$79.65_{\pm0.61}$	**$81.38_{\pm0.56}$**
	40%	$71.62_{\pm0.40}$	**$79.46_{\pm0.93}$**	$71.81_{\pm0.72}$	**$79.11_{\pm1.00}$**	$72.77_{\pm0.48}$	**$79.86_{\pm0.86}$**	$74.37_{\pm0.67}$	**$79.64_{\pm1.14}$**
Citeseer	10%	$71.08_{\pm0.63}$	**$73.95_{\pm0.59}$**	$71.62_{\pm0.79}$	**$73.73_{\pm0.58}$**	$72.74_{\pm0.46}$	**$73.85_{\pm1.07}$**	$71.15_{\pm0.75}$	**$73.41_{\pm0.64}$**
	20%	$68.79_{\pm0.49}$	**$73.32_{\pm0.58}$**	$69.68_{\pm0.59}$	**$73.26_{\pm0.45}$**	$70.67_{\pm0.33}$	**$73.61_{\pm0.69}$**	$70.67_{\pm0.53}$	**$72.66_{\pm0.73}$**
	40%	$63.98_{\pm1.02}$	**$73.35_{\pm0.50}$**	$64.81_{\pm0.87}$	**$73.07_{\pm0.71}$**	$65.46_{\pm0.55}$	**$73.64_{\pm1.02}$**	$68.53_{\pm1.80}$	**$72.80_{\pm0.63}$**
Pubmed	10%	$82.86_{\pm0.17}$	**$85.14_{\pm0.22}$**	$81.85_{\pm0.17}$	**$83.96_{\pm0.22}$**	$83.48_{\pm0.12}$	**$85.24_{\pm0.15}$**	$86.60_{\pm0.07}$	**$86.92_{\pm0.13}$**
	20%	$80.11_{\pm0.16}$	**$84.96_{\pm0.19}$**	$78.92_{\pm0.20}$	**$84.33_{\pm0.28}$**	$80.55_{\pm0.11}$	**$85.06_{\pm0.20}$**	$85.98_{\pm0.08}$	**$86.66_{\pm0.18}$**
	40%	$74.45_{\pm0.19}$	**$84.89_{\pm0.21}$**	$72.47_{\pm0.24}$	**$83.98_{\pm0.17}$**	$74.78_{\pm0.18}$	**$84.73_{\pm0.25}$**	$85.36_{\pm0.09}$	**$86.42_{\pm0.11}$**
Arxiv	10%	$64.23_{\pm0.10}$	**$68.15_{\pm0.89}$**	$62.39_{\pm0.28}$	**$67.96_{\pm1.05}$**	$62.92_{\pm0.27}$	**$67.99_{\pm1.00}$**	$62.73_{\pm0.27}$	**$66.77_{\pm1.51}$**
	20%	$62.35_{\pm0.17}$	**$68.21_{\pm1.01}$**	$59.60_{\pm0.40}$	**$67.84_{\pm1.10}$**	$60.58_{\pm0.27}$	**$67.95_{\pm1.11}$**	$60.52_{\pm0.25}$	**$66.56_{\pm1.19}$**
	40%	$57.55_{\pm0.14}$	**$68.73_{\pm0.32}$**	$54.86_{\pm0.38}$	**$68.52_{\pm0.53}$**	$56.33_{\pm0.21}$	**$68.47_{\pm0.40}$**	$56.38_{\pm0.23}$	**$67.37_{\pm1.64}$**
Products	10%	$76.05_{\pm0.16}$	**$77.66_{\pm0.22}$**	$75.12_{\pm0.28}$	**$76.16_{\pm0.31}$**	$74.78_{\pm0.33}$	**$76.41_{\pm0.43}$**	$71.62_{\pm0.61}$	**$74.20_{\pm0.41}$**
	20%	$72.48_{\pm0.21}$	**$77.15_{\pm0.62}$**	$72.03_{\pm0.37}$	**$75.62_{\pm1.06}$**	$71.71_{\pm0.27}$	**$76.12_{\pm0.59}$**	$68.03_{\pm0.44}$	**$74.57_{\pm1.00}$**
	40%	$66.26_{\pm0.37}$	**$77.02_{\pm0.56}$**	$65.52_{\pm0.62}$	**$76.20_{\pm0.40}$**	$64.70_{\pm0.31}$	**$75.85_{\pm0.46}$**	$60.41_{\pm0.57}$	**$75.01_{\pm0.64}$**

Table 4: Node classification accuracy (% ± σ) under non-targeted attack (DICE). The best results are in bold. OOT means that the result could not be obtained in 15 days.

Dataset	Ptb Rate	Pro-GNN	STABLE	HANG-quad	GraphEdit	LLM4RGNN
Cora	0%	$80.85_{\pm0.44}$	**$85.09_{\pm0.21}$**	$79.41_{\pm0.55}$	$76.35_{\pm0.38}$	$84.13_{\pm0.33}$
	10%	$81.56_{\pm0.36}$	$81.33_{\pm0.72}$	$78.26_{\pm0.29}$	$73.73_{\pm0.57}$	**$81.84_{\pm0.71}$**
	20%	$78.32_{\pm0.33}$	$78.80_{\pm0.83}$	$76.45_{\pm0.50}$	$69.86_{\pm0.71}$	**$80.96_{\pm0.61}$**
	40%	$71.76_{\pm0.34}$	$76.72_{\pm0.93}$	$74.28_{\pm0.42}$	$66.95_{\pm0.48}$	**$79.46_{\pm0.93}$**
Citeseer	0%	$71.11_{\pm0.45}$	$72.51_{\pm1.37}$	$71.18_{\pm0.64}$	$72.60_{\pm0.57}$	**$74.20_{\pm0.56}$**
	10%	$70.81_{\pm0.68}$	$70.27_{\pm1.62}$	$70.29_{\pm0.70}$	$68.99_{\pm0.52}$	**$73.95_{\pm0.59}$**
	20%	$69.21_{\pm0.79}$	$69.11_{\pm1.57}$	$70.05_{\pm0.77}$	$69.31_{\pm0.41}$	**$73.32_{\pm0.58}$**
	40%	$68.04_{\pm1.38}$	$67.55_{\pm0.86}$	$68.79_{\pm1.09}$	$66.59_{\pm0.74}$	**$73.35_{\pm0.50}$**
Pubmed	0%	OOT	$84.07_{\pm0.17}$	$84.98_{\pm0.13}$	$85.38_{\pm0.07}$	**$86.21_{\pm0.13}$**
	10%	OOT	$81.68_{\pm0.21}$	$84.97_{\pm0.15}$	$82.12_{\pm0.07}$	**$85.14_{\pm0.22}$**
	20%	OOT	$79.08_{\pm0.23}$	$84.94_{\pm0.18}$	$81.11_{\pm0.10}$	**$84.96_{\pm0.19}$**
	40%	OOT	$74.51_{\pm0.45}$	$84.86_{\pm0.20}$	$83.04_{\pm0.09}$	**$84.89_{\pm0.21}$**
Arxiv	0%	OOT	$66.70_{\pm0.19}$	$67.85_{\pm0.17}$	-	**$68.16_{\pm0.36}$**
	10%	OOT	$64.93_{\pm0.19}$	$65.99_{\pm0.19}$	-	**$68.15_{\pm0.89}$**
	20%	OOT	$62.64_{\pm0.44}$	$63.74_{\pm0.23}$	-	**$68.21_{\pm1.01}$**
	40%	OOT	$58.55_{\pm0.37}$	$59.23_{\pm0.17}$	-	**$68.73_{\pm0.32}$**
Products	0%	OOT	$79.23_{\pm0.33}$	**$80.61_{\pm0.18}$**	-	$79.04_{\pm0.42}$
	10%	OOT	$76.72_{\pm0.32}$	$77.06_{\pm0.29}$	-	**$77.66_{\pm0.22}$**
	20%	OOT	$75.15_{\pm0.27}$	$75.52_{\pm0.24}$	-	**$77.15_{\pm0.62}$**
	40%	OOT	$73.68_{\pm0.26}$	$69.01_{\pm0.17}$	-	**$77.02_{\pm0.56}$**

Main Result Against Nettack

Targeted

Table 5: Node classification accuracy (% ± σ) under targeted attack (Nettack). Bolded results indicate improved performance.

Dataset		GCN		GAT		RGCN		Sim-PGCN	
	Ptb Num	Vanilla	LLM4RGNN	Vanilla	LLM4RGNN	Vanilla	LLM4RGNN	Vanilla	LLM4RGNN
Cora	0	$85.92_{\pm 2.21}$	$\mathbf{87.50}_{\pm 1.79}$	$85.66_{\pm 1.90}$	$\mathbf{87.11}_{\pm 1.64}$	$86.45_{\pm 1.87}$	$\mathbf{88.55}_{\pm 2.57}$	$86.97_{\pm 1.09}$	$86.71_{\pm 2.39}$
	1	$79.87_{\pm 2.50}$	$\mathbf{85.79}_{\pm 1.75}$	$82.11_{\pm 2.06}$	$\mathbf{86.71}_{\pm 2.24}$	$82.50_{\pm 1.87}$	$\mathbf{87.37}_{\pm 2.64}$	$81.71_{\pm 2.39}$	$\mathbf{86.05}_{\pm 2.37}$
	2	$76.32_{\pm 1.77}$	$\mathbf{85.66}_{\pm 2.08}$	$75.13_{\pm 4.42}$	$\mathbf{86.18}_{\pm 1.79}$	$77.37_{\pm 1.29}$	$\mathbf{87.37}_{\pm 1.58}$	$77.76_{\pm 1.37}$	$\mathbf{84.34}_{\pm 1.99}$
	3	$71.45_{\pm 2.43}$	$\mathbf{84.47}_{\pm 1.15}$	$70.79_{\pm 3.85}$	$\mathbf{85.00}_{\pm 2.58}$	$72.11_{\pm 1.64}$	$\mathbf{86.32}_{\pm 2.44}$	$71.97_{\pm 2.89}$	$\mathbf{85.39}_{\pm 1.37}$
	4	$64.47_{\pm 3.38}$	$\mathbf{84.61}_{\pm 2.04}$	$65.79_{\pm 3.86}$	$\mathbf{85.39}_{\pm 1.99}$	$67.24_{\pm 1.90}$	$\mathbf{86.84}_{\pm 1.95}$	$63.55_{\pm 3.58}$	$\mathbf{84.61}_{\pm 2.50}$
	5	$55.13_{\pm 4.01}$	$\mathbf{84.08}_{\pm 1.61}$	$60.00_{\pm 4.68}$	$\mathbf{83.03}_{\pm 1.90}$	$60.26_{\pm 2.48}$	$\mathbf{85.13}_{\pm 1.56}$	$52.63_{\pm 4.40}$	$\mathbf{83.55}_{\pm 1.69}$
Citeseer	0	$86.31_{\pm 1.28}$	$\mathbf{86.77}_{\pm 1.02}$	$86.92_{\pm 0.77}$	$\mathbf{87.38}_{\pm 1.02}$	$85.23_{\pm 1.41}$	$\mathbf{86.92}_{\pm 1.72}$	$80.92_{\pm 2.50}$	$\mathbf{84.62}_{\pm 2.84}$
	1	$84.62_{\pm 2.57}$	$\mathbf{86.46}_{\pm 0.62}$	$85.54_{\pm 1.23}$	$\mathbf{86.92}_{\pm 1.03}$	$82.15_{\pm 1.85}$	$\mathbf{86.31}_{\pm 1.45}$	$80.00_{\pm 3.30}$	$\mathbf{84.31}_{\pm 2.36}$
	2	$81.69_{\pm 2.62}$	$\mathbf{85.38}_{\pm 1.03}$	$83.38_{\pm 1.79}$	$\mathbf{86.38}_{\pm 1.61}$	$74.46_{\pm 4.25}$	$\mathbf{85.85}_{\pm 1.51}$	$79.54_{\pm 4.30}$	$\mathbf{84.62}_{\pm 1.54}$
	3	$79.08_{\pm 3.24}$	$\mathbf{86.00}_{\pm 0.63}$	$81.38_{\pm 3.11}$	$\mathbf{85.38}_{\pm 1.42}$	$72.00_{\pm 3.94}$	$\mathbf{85.54}_{\pm 1.41}$	$76.62_{\pm 5.99}$	$\mathbf{83.85}_{\pm 2.31}$
	4	$76.77_{\pm 5.26}$	$\mathbf{86.00}_{\pm 0.83}$	$77.08_{\pm 4.54}$	$\mathbf{85.69}_{\pm 1.55}$	$69.08_{\pm 2.70}$	$\mathbf{85.38}_{\pm 1.58}$	$73.23_{\pm 7.51}$	$\mathbf{84.00}_{\pm 1.41}$
	5	$74.31_{\pm 5.29}$	$\mathbf{86.00}_{\pm 1.08}$	$73.69_{\pm 8.44}$	$\mathbf{85.54}_{\pm 1.02}$	$64.31_{\pm 3.94}$	$\mathbf{84.46}_{\pm 1.45}$	$68.31_{\pm 10.62}$	$\mathbf{82.46}_{\pm 2.20}$

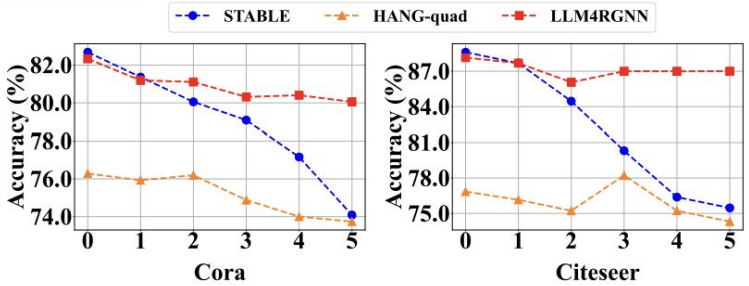


Figure 3: Node classification accuracy under targeted attack (Nettack) with different perturbations.

Main Result Against Adaptive Attack

Targeted (worst case scenario)

Table 6: Node classification accuracy of Pubmed under adaptive attack. Bolded results indicate improved performance.

Ptb Rate	GCN		RGCN		SimP-GCN	
	Vanilla	LLM4RGNN	Vanilla	LLM4RGNN	Vanilla	LLM4RGNN
0%	$85.62_{\pm 0.10}$	$\mathbf{86.21}_{\pm 0.13}$	$85.84_{\pm 0.10}$	$\mathbf{86.35}_{\pm 0.10}$	$87.26_{\pm 0.08}$	$\mathbf{87.53}_{\pm 0.14}$
5%	$84.23_{\pm 0.12}$	$\mathbf{84.70}_{\pm 0.28}$	$84.49_{\pm 0.10}$	$\mathbf{85.43}_{\pm 0.20}$	$86.21_{\pm 0.10}$	$\mathbf{86.36}_{\pm 0.25}$
10%	$81.51_{\pm 0.20}$	$\mathbf{84.82}_{\pm 0.26}$	$82.69_{\pm 0.21}$	$\mathbf{85.58}_{\pm 0.20}$	$85.86_{\pm 0.16}$	$\mathbf{86.11}_{\pm 0.16}$

Model Analysis

- Ablation Study under Mettack with a 20% perturbation rate, where GCN is selected as training GNN
- Across all settings, LLM4RGNN Full consistently outperforms other settings.

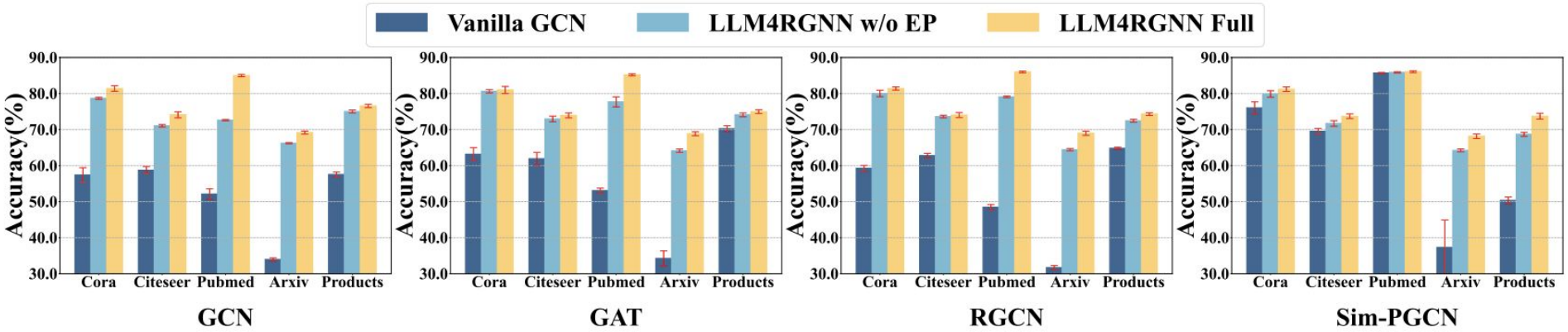


Figure 5: Performance comparison in different settings against Mettack with a 20% perturbation rate.

Model Analysis Cont.

- Impact of Text Quality on model's effectiveness since LLM4RGNN relies on textual information for reasoning.
- LLM4RGNN's performance varies by less than 2 points under 10-40% perturbations and surpasses existing robust GNN frameworks.

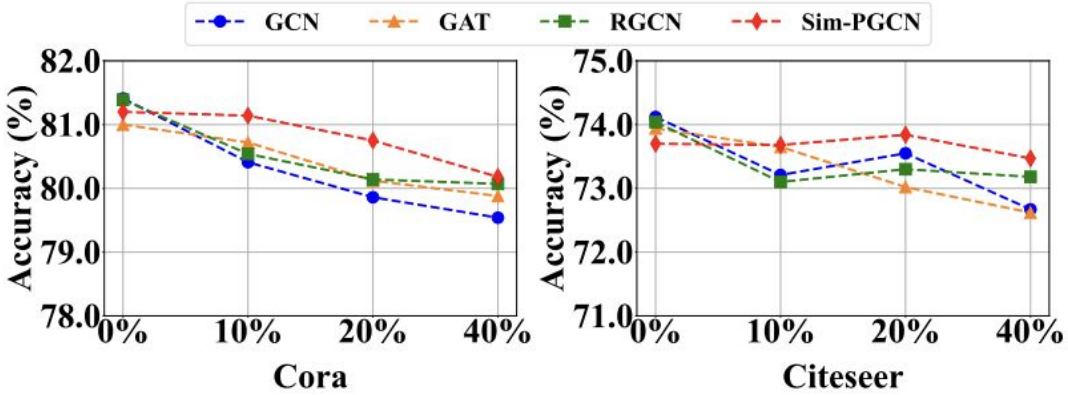


Figure 4: Node classification accuracy of LLM4RGNN under Mettack-20% with different text perturbations.

Model Analysis Cont.

- Comparison with other LLMs (Llama2-7B, Llama2-13B, Llama3-8B, and Mistral-7B).
- Also used closed source GPT-3.5 and GPT-4.
- Local LLMs are significantly superior to GPT-3.5.
- A stronger LLM yields better overall performance.

Table 7: Performance comparison with different LLMs against Mettack with a 20% perturbation rate.

LLM		Cora			Citeseer		
		AdvEdge (↓)	ACC (↑) w/o EP	ACC (↑) Full	AdvEdge (↓)	ACC (↑) w/o EP	ACC (↑) Full
Close source	GPT-3.5	186(3.52%)	$73.56_{\pm 0.49}$	$75.36_{\pm 1.73}$	231(5.47%)	$66.69_{\pm 1.04}$	$68.92_{\pm 1.63}$
	GPT-4	**90(1.71%)**	$\mathbf{79.83_{\pm 0.31}}$	$\mathbf{81.76_{\pm 0.87}}$	**73(1.73%)**	$\mathbf{71.04_{\pm 0.57}}$	$\mathbf{72.81_{\pm 0.60}}$
Open source	Llama2-7B	156(2.96%)	$78.76_{\pm 0.42}$	$80.99_{\pm 0.78}$	203(4.80%)	$69.83_{\pm 0.78}$	$72.86_{\pm 0.93}$
	Llama2-13B	132(2.50%)	$79.00_{\pm 0.30}$	$81.35_{\pm 1.15}$	93(2.20%)	$70.81_{\pm 0.73}$	$72.84_{\pm 0.85}$
	Llama3-8B	107(2.03%)	$\mathbf{79.22_{\pm 0.34}}$	$\mathbf{81.71_{\pm 0.51}}$	**91(2.15%)**	$70.85_{\pm 0.64}$	$74.06_{\pm 0.78}$
	Mistral-7B	**102(1.93%)**	$78.68_{\pm 0.28}$	$81.41_{\pm 0.77}$	92(2.18%)	$\mathbf{71.07_{\pm 0.35}}$	$\mathbf{74.12_{\pm 0.85}}$

Weaknesses

- Limited to text-attributed graphs (authors noted a plan to extend to graphs without text)
- Supervision depends on GPT-4 outputs (biases/errors)
- Only covers 4 types of attack methods

Conclusion

- LLM4RGNN is an LLM-based robust graph structure inference framework, which distills the inference capability of GPT-4 into a local LLM for identifying malicious edges and an LM-based edge predictor for finding missing important edges, to efficiently purify attacked graph structure, making GNNs more robust.
- Experiments demonstrate that LLM4RGNN significantly improves various GNN’s robustness
- In the future, LLM4RGNN could be extended to graphs without text.